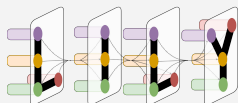


Graph-based Neural Multi-Doc Summarization

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Yale - .github.io

Text Summarization

The process of **shortening** a text to create a summary with the **major points**

- Single-doc summarization
- Multi-doc summarization (MDS):
A cluster of documents about the same topic

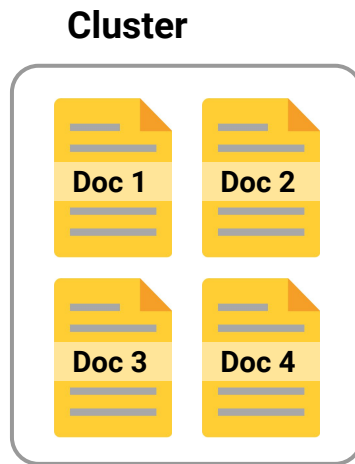
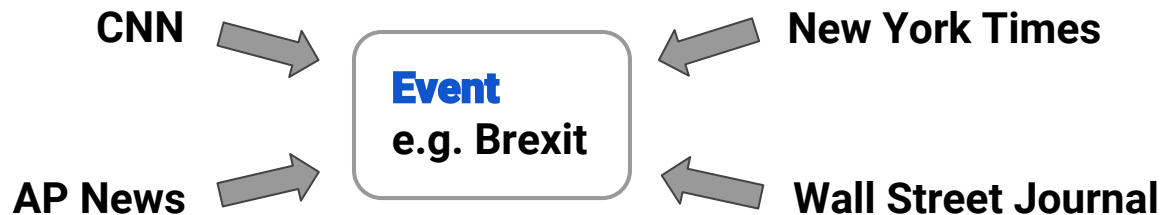
Methods

- Extractive:
Extract salient sentences (phrases) from the original text
- Abstractive:
Generate a summary from scratch, involving paraphrasing

Why Multi-Doc Summarization (MDS)?

Often times, we want a summary for a whole topic, rather than one document

- E.g. different news articles about the same event



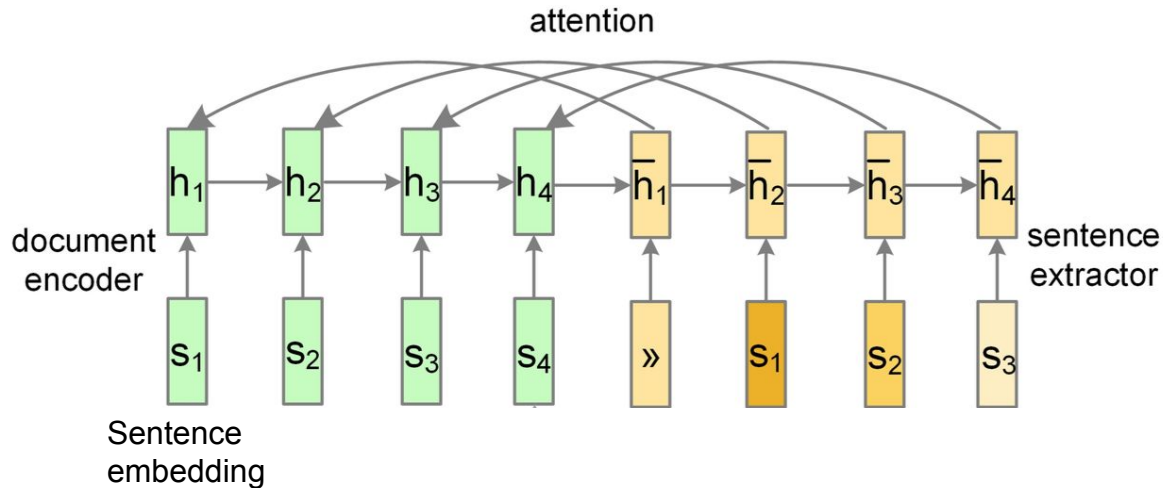
- More challenging, as we need to think about the relationships between documents

Recent Models for Single-Doc Summarization

RNN encoder-decoder model

- E.g. Cheng and Lapata (2016),

See et al. (2017) - abstractive

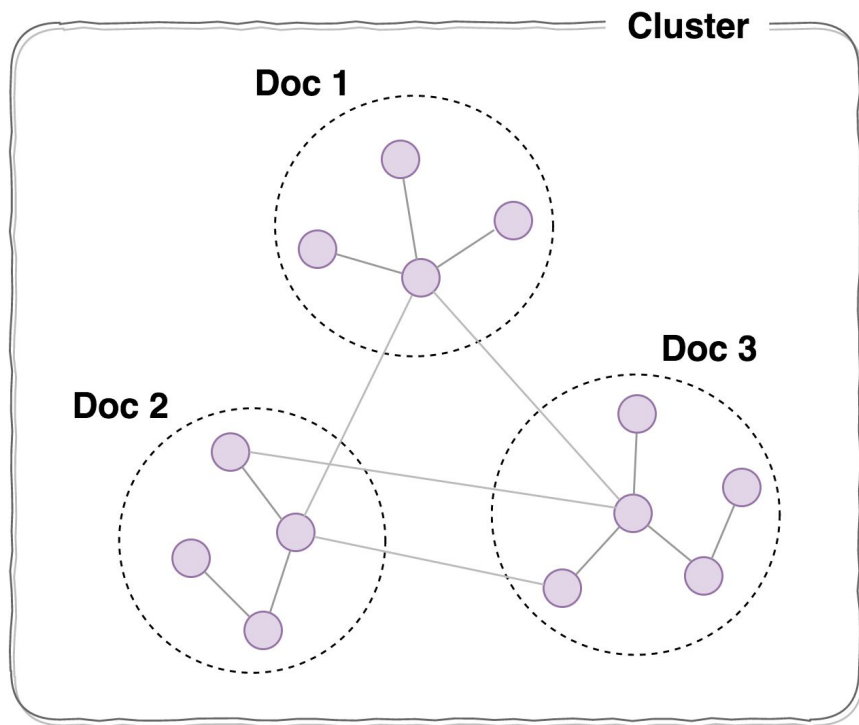


Can we extend this to Multi-Docs?

Several issues:

- RNN encoder
 - A simple RNN cannot capture the relationships between different documents
 - Christensen et al. (2013) demonstrates the importance of considering **sentence relations among multiple documents**
- RNN decoder
 - Requires large training data, which MDS datasets are lacking in
 - Instead, we break down the summarization task into **sentence salience estimation** and **sentence selection**

Graph Representation of A Document Cluster



Sentence relation graph

- node = sentence
- edge / weight
= relation between two sentences

Graph Representation Methods

- **Cosine Similarity Graph:**
 - LexRank (Erkan and Radev, 2004)
 - Edge weight: tf-idf cosine similarity between sentences
- **Approximate Discourse Graph (ADG):**
 - Christensen et al. (2013)
- **Personalized Discourse Graph (PDG):**
 - Our modification of ADG

Approximate Discourse Graph (ADG)

- Christensen et al. (2013)
- Construct edges between sentences by counting **discourse relation indicators**:
 - deverbial noun references, event and entity continuations, discourse markers, co-referent mentions, etc.
- Edge weight is discrete & not diverse
 - Often 0, 1, or 2

Personalized Discourse Graph (PDG)

- Modification of ADG: **Diversify edge weights** via sentence personalization

- Sentence personalization score:

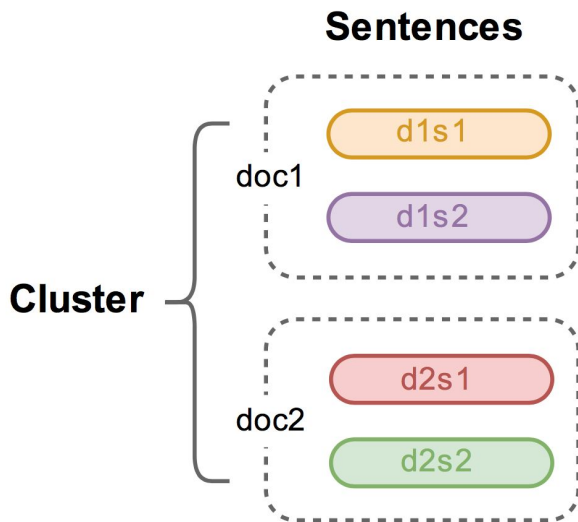
- Consider **weighting for each sentence**
- Computed by simple macro-level features (sentence position, length, etc.) (Christensen et al. 2013)

- Edge weight is transformed via
$$w_{PDG}(u, v) = \frac{w_{ADG}(u, v)s(u)}{\sum_{u' \in V} w_{ADG}(u', v)s(u')}$$

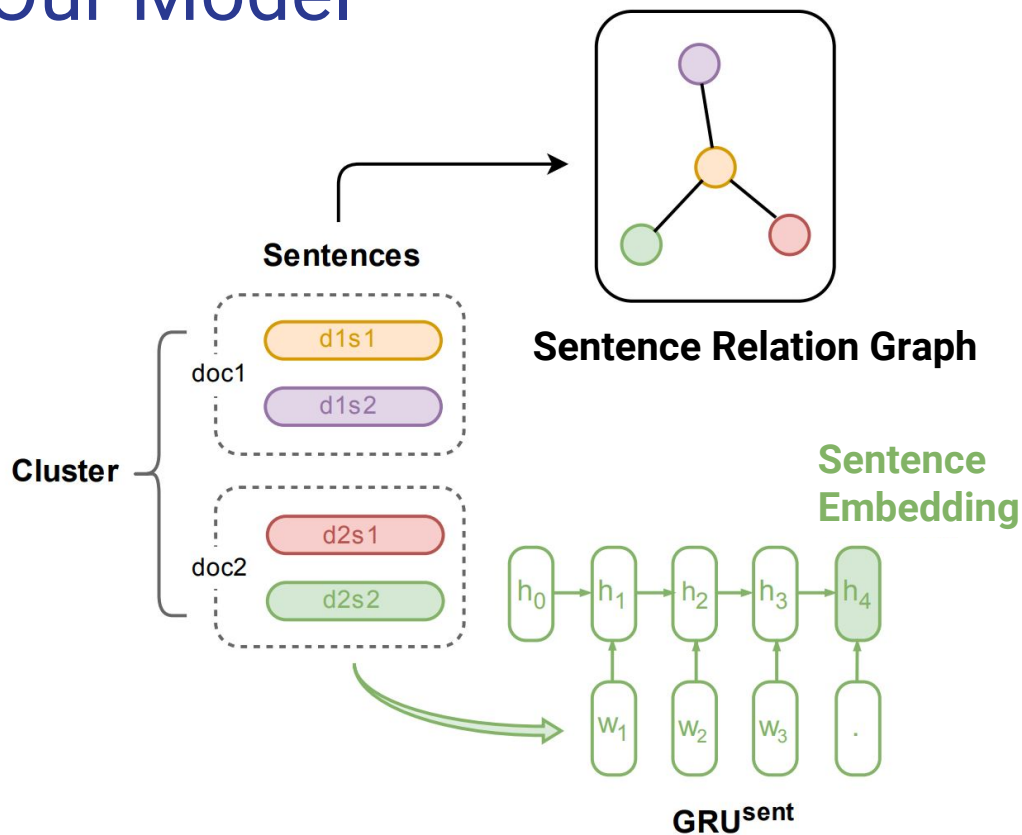
where $s(\cdot)$ is the personalization scores

Our Model

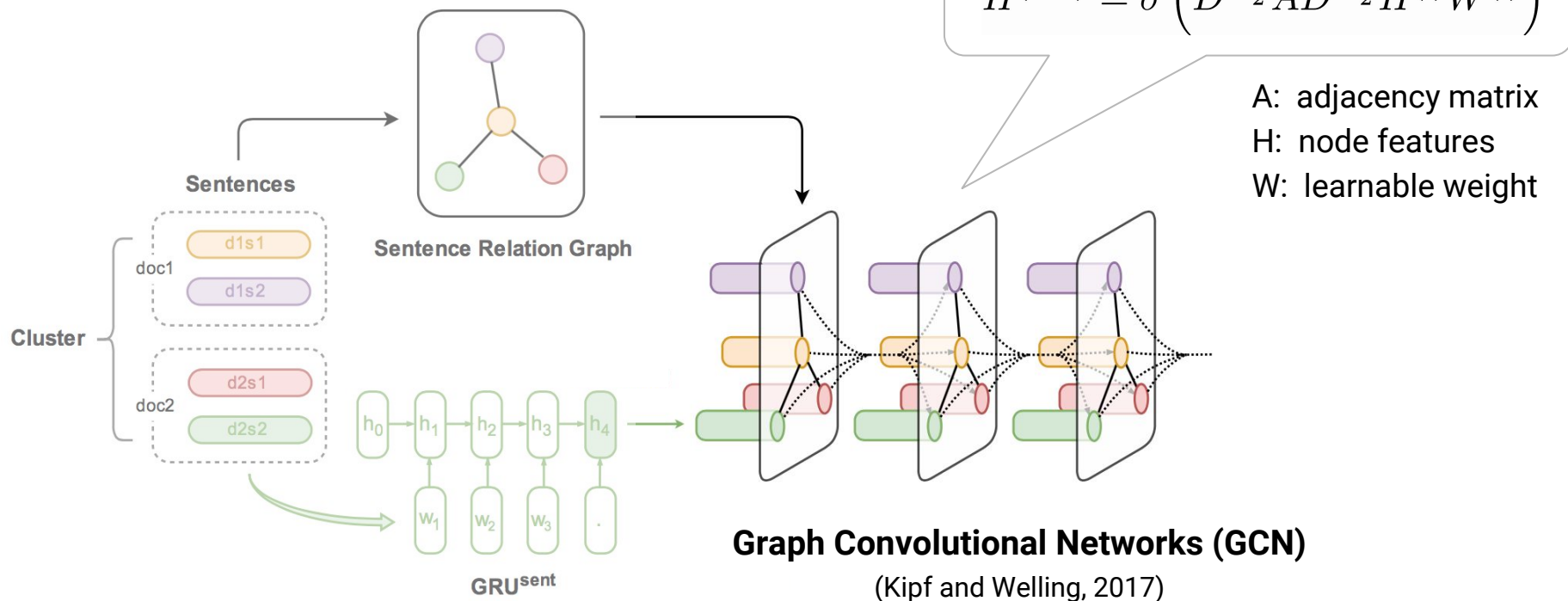
- **Neural Net** for powerful representation learning
- **Graph** for capturing the sentence relations in a document cluster



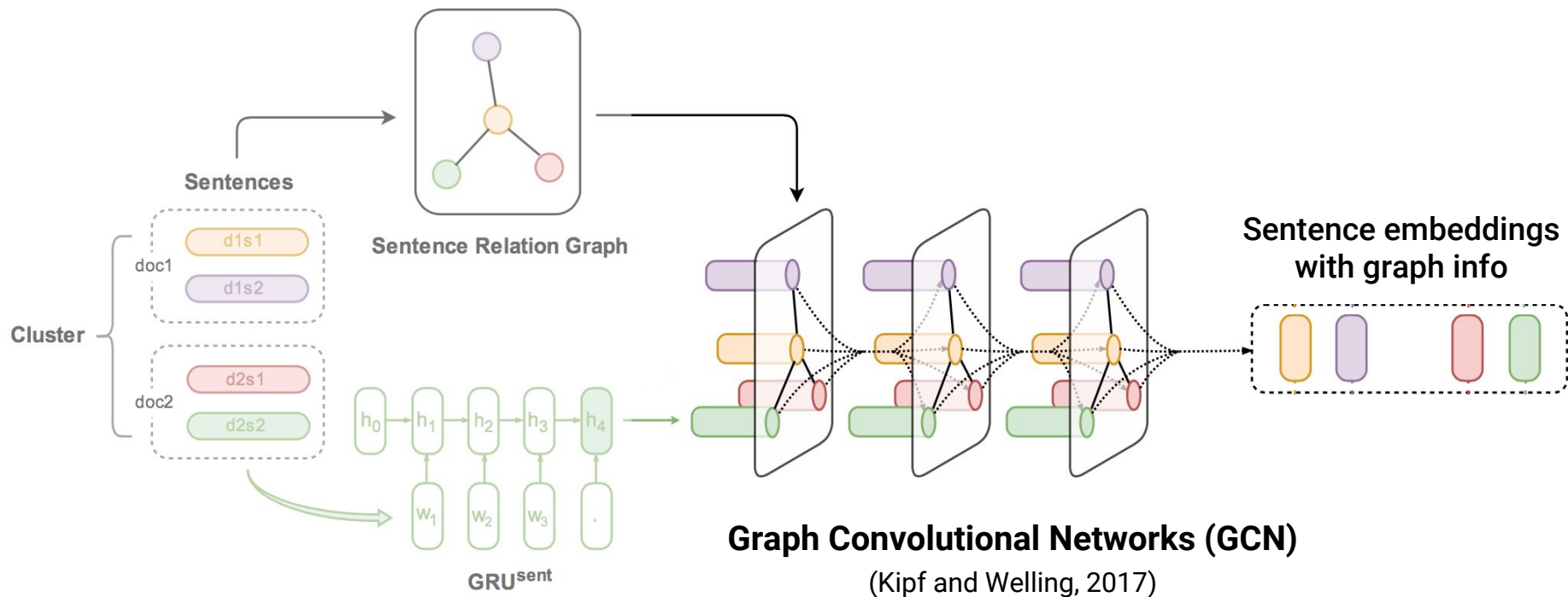
Our Model



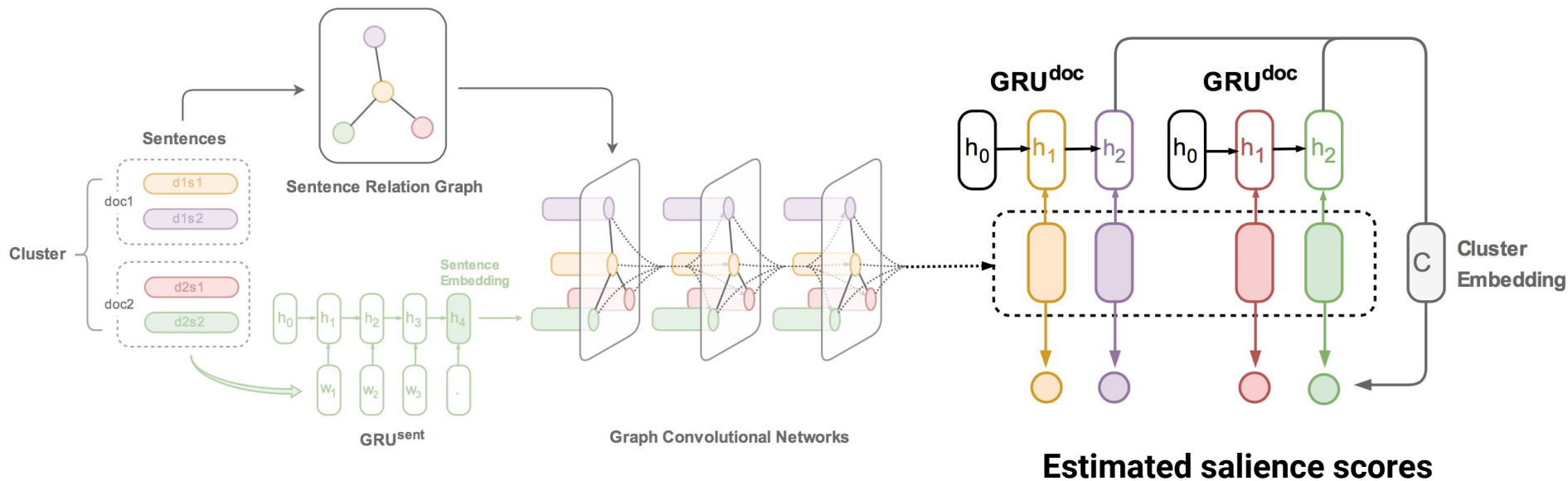
Our Model - GCN



Our Model - GCN



Our Model - Saliency Estimation



Training & Testing

Training:

- Obtain target salience score for each sentence:
Average of the ROUGE 1 & 2 scores
- Minimize the loss between the **estimated** scores and the **target** scores

Testing:

- Select sentences with high predicted scores in a **greedy manner**
- Avoid redundancy during selection:
Add the sentence if its cosine similarity to current summary is < 0.5
- (Hong & Nenkova, 2014)

Experiments - Data set

Data set: DUC MDS-task (Train: 2001 & 2002, Dev: 2003, Test: 2004)

	DUC'01	DUC'02	DUC'03	DUC'04
# of Clusters	30	59	30	50
# of Documents	309	567	298	500
# of Sentences	24498	16090	7721	13270
Vocabulary Size	28188	22174	13248	18036
Summary Length	100 words	100 words	100 words	665 Bytes

Experiments - Evaluation

Metric: ROUGE-1, 2 (Lin, 2004)

Model comparison

Baseline: vanilla GRU (NO graph)

Graph-based neural:

- GRU + GCN (cosine similarity graph)

- GRU + GCN (ADG)

- GRU + GCN (PDG)

Research Questions

- Does the GCN model with sentence relation graphs improve upon the vanilla RNN model?
- How does our model compare with traditional graph-based summarizers?
- Comparison of different sentence relation graphs:
 - Cosine Similarity Graph
 - ADG
 - PDG

Results

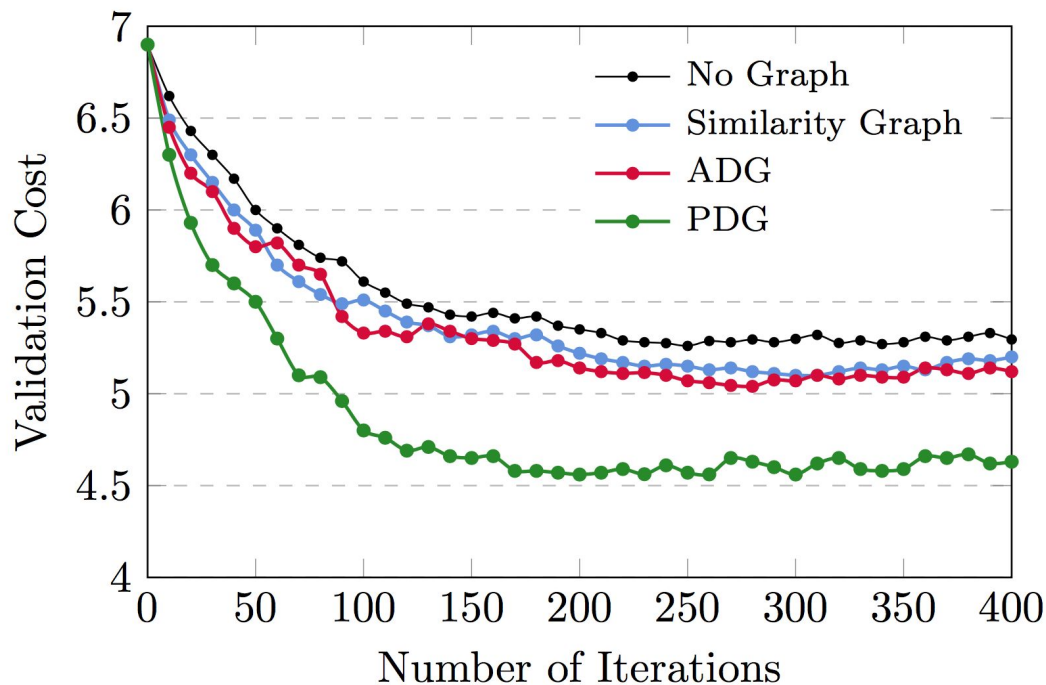
Our models

	ROUGE-1	ROUGE-2
GRU only (No Graph)	36.04	8.47
GCN: Similarity Graph	37.33	8.78
GCN: ADG	37.41	8.97
GCN: PDG	38.23	9.48

Other baseline & SOTA multi-doc summarizers

	ROUGE-1	ROUGE-2
Centroid (Radev et al., 2007)	36.41	7.97
CLASSY 04 (Conroy et al., 2007)	37.62	8.96
LexRank (Erkan and Radev, 2004)	35.95	7.47
G-Flow (Christensen et al., 2013)	35.30	8.27
SVR (Li et al., 2007)	36.18	9.34
RegSum (Hong & Nenkova, 2014)	38.57	9.75

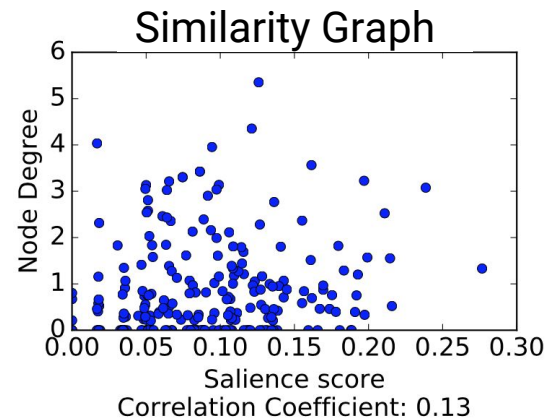
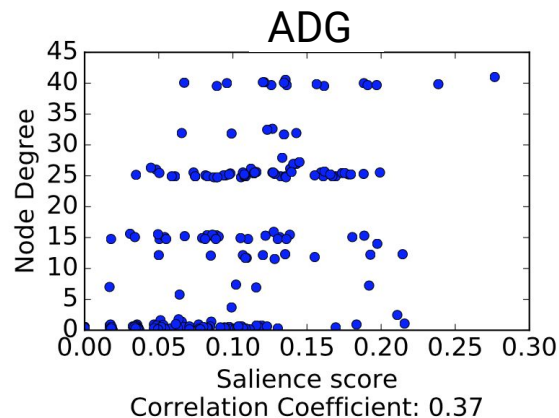
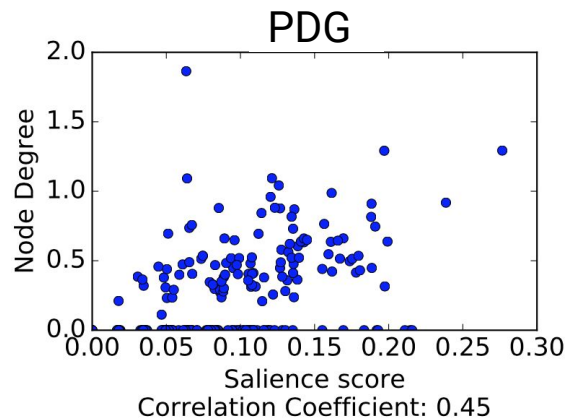
Analysis - Learning Curve



PDG

- converges the fastest
- shows the best generalization

Analysis - Graph Characteristics



x axis = saliency, y axis = node degree

PDG:	stronger correlation, thanks to the improvement in edge weight
ADG:	discrete weight
Similarity Graph:	weaker correlation

Conclusion & Contribution

Our GCN model performed ...

1. better than vanilla GRU
=> Graph captures **sentence relations across documents**
2. better than traditional graph-based summarizers
=> Powerful representation learning by neural net

Gateway to incorporating graph-based techniques into neural summarization



Conclusion & Contribution

Our **Personalized Discourse Graph (PDG)** provides the most informative guide to salience estimation, among the three graph representation methods.

- => Advantage of **discourse relations** over simple similarity
- Efficacy of adjusting edge weights via **sentence personalization**

Thank you!

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